

3-Month Delivery Roadmap

Milestones and weekly workstreams for the new team across the three components — **Event Processor**, **Aggregator**, and **Monetization Engine** — plus the internal **Simulator**. Scoped for incremental, demoable progress rather than a big-bang build.

HORIZON 12 weeks · 3 milestones CADENCE Weekly demo + progress

REPOS EP · Aggregator · ME ASSIST Devin (scoped tasks)

M1 WEEKS 1–4

Foundations & I/O Primitives

Repos, CI/CD, common libs, Mongo schema & indexes, and the core Kafka/HPOS building blocks — proven with the Simulator.

EXIT DEMO

Sample events flow Kafka → HPOS via the primitives; Mongo collections + indexes live.

M2 WEEKS 5–8

Event Processor & Aggregator Core

EP writes RAW_EVENTS direct-to-HPOS with resiliency; Aggregator scheduler, validation suite, classification and enrichment.

EXIT DEMO

A day of raw events → validated, classified →

M3 WEEKS 9–12

Monetization Engine & End-to-End

ME generates & validates the ACO file, NDM delivery with retries, rejected/late handling — then full E2E and load hardening.

EXIT DEMO

End-to-end monthly ACO file generated, validated and NDM-delivered via the Simulator.

written to Mongo +
Parquet.

WEEKLY WORKSTREAMS

Bars show the active weeks per workstream. Dashed lines mark the M1 / M2 / M3 boundaries.



Platform / Cross-cutting Simulator Event Processor Aggregator

TASK BACKLOG BY COMPONENT

Your listed tasks plus additions derived from the three LLDs. Well-scoped items (validations, primitives, simulator flows) are good candidates to hand to Devin with human review.

Simulator · internal test

component

- ☐ Publish sample **success billing events** to the Kafka topic
- ☐ Publish **negative/edge samples** — missing mandatory fields, bad date format, unknown aggregator/cluster
- ☐ Consume from Kafka → write **RAW_EVENTS to HPOS**
- ☐ Retrieve events from Kafka (consumer harness)
- ☐ Configurable **volume/throughput** to simulate 2B/month load
- ☐ **Late-arrival** event generator (event-time in a past window)
- ☐ Duplicate-eventId **idempotency/replay** harness
- ☐ **Master-data seeder** — bank details & mappings for enrichment tests

Event Processor

- ☐ Kafka consumer + **Consumer Group** config (single group, partition ownership)
- ☐ Retrieve events from Kafka
- ☐ **Direct-to-HPOS** write of RAW_EVENTS (as-is JSON, deterministic key by eventId)
- ☐ **Offset commit after successful write**
- ☐ HPOS resiliency — retry/backoff, circuit-breaker **pause/resume**
- ☐ Partition & parallelism config; producer UUID-key guidance
- ☐ Metrics — consumer lag, write latency, paused duration

Aggregator

- ☐ Scheduler with **grace window** (6h, configurable)
- ☐ **Bounded batch read** from RAW_EVENTS (paginated)
- ☐ Schema validation · Mandatory-field validation

Monetization Engine

- ☐ Monthly **scheduler** (1st–5th, last month)
- ☐ Run modes — monthly / on-demand targeted dates / late-events (delta)
- ☐ Read **Monthly Aggregated** collection

- ☐ **Attribute data-format** validation (e.g. date format)
- ☐ Aggregator validation · Cluster validation
- ☐ Billing rules → **Billable / Non-Billable / Rejected**
- ☐ **Fetch bank details by aggregator** + mapping (Source Id / EVENTID / Cluster Id)
- ☐ Mongo **bulk upsert by eventId** (3 collections)
- ☐ Parquet write (10k/file) to 3 prefixes
- ☐ **Delete RAW after both acks** (batch)
- ☐ **Late-arrival** handling (write to event-time hour)
- ☐ Concurrency (lease/lock per prefix) + quarantine for poison objects

- ☐ **Enrich** bank + mapping master data
- ☐ Pricing / billing rules + **ACO field mapping**
- ☐ **ACO file generation** (deterministic name)
- ☐ **Pre-send validation gate** (+ cross-check vs master)
- ☐ Validation-fail handling (hold, alert, VALIDATION_FAILED)
- ☐ Rejected records (missing master) → exceptions collection + delta
- ☐ **NDM transfer** + retries/backoff + alert
- ☐ Status lifecycle + **atomic claim** (idempotency)

Platform & Cross-cutting

- ☐ Repos + CI/CD pipelines (EP · Aggregator · ME) + branch strategy
- ☐ **Mongo schema finalization & creation**
- ☐ Config management + secrets
- ☐ Environments (dev/test) + infra-as-code
- ☐ Security review — bank-data handling
- ☐ Shared event model / common libraries
- ☐ **Identify & apply Mongo indexes**
- ☐ Observability — logging, metrics, dashboards, alerts
- ☐ Load & resiliency testing (2B/month target)
- ☐ Runbooks + on-call / alerting playbook

SUGGESTED REPORTING CADENCE

How to show weekly progress

Weekly demo (Fri)

A working slice each week — even a primitive or a Simulator flow. Progress you can point at, not status text.

Per-workstream burn-up

Track % complete against the backlog per component; the Gantt above is your weekly RAG view.

Milestone exit review

At W4 / W8 / W12, walk the exit demo & criteria with your manager before starting the next milestone.

Metering Engine — 3-Month Delivery Roadmap · indicative; sequence & durations to be confirmed with the team.